
Command Line Interface and DOS Experience

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1. Introduction

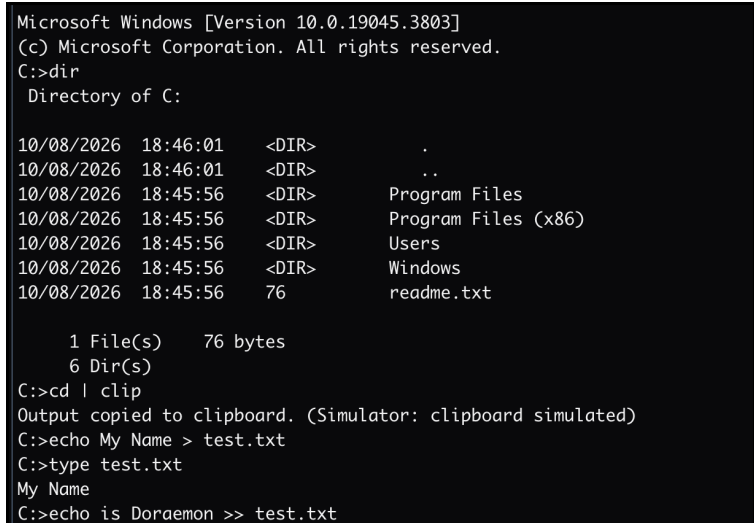
- Purpose of using Command Line Interface:

The Command Line interface (CLI) provides a faster way to relay instructions to the computer as opposed to clicking through a Graphical User Interface (GUI). This makes it much easier and less time intensive to perform repetitive tasks and automate some other kinds of operations.

1.1 Commands shown in Class:

```
# Win + R -> cmd
dir
#Listings special: cd . -> Current folder, cd . . -> Parent
  ↪ folder
#3rd fn of cd -> Shows name of current folder
mkdir foldername
#creates a new directory
rmdir foldername
#deletes a directory (has to be empty unless you force it)
move file_name1 destination
#moves a file to another location, same syntax logic as copy
ren old_name new_name
#renames a file
del file_name
#deletes a file
cd | clip
#piping, copying the folder name to clipboard
start .
#opens folder in which you are in
/?
#help
#no touch in Windows, directly echo smth to get the file out
  ↪ of nowhere
>
#overwrite
>>
#append
type file_name
#views file in cmd
copy file_name1 file_name2
```

```
#even when second file DNE
copy ..\a\*.txt ..\c
#everything in one line when you are in folder b
/p
#pauses output aka asks y/n every step e.g., deletion
/s
#forces, force del like when you have smth in the dir you want
  ↳ to remove
wd -d .
#go to the designated folder in GUI and type this on the
  ↳ window to directly access Powershell
cmd
#same way, cmd also works and cmd is ultimate destination even
  ↳ from Powershell
copy /filepath .
#copies to this folder without creation of new name
magick clipboard: cli.png
#first use imagemagick dwld and then use snipping tool
cls
#clears the clutter on screen
```



```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>dir
Directory of C:

10/08/2026  18:46:01    <DIR>        .
10/08/2026  18:46:01    <DIR>        ..
10/08/2026  18:45:56    <DIR>        Program Files
10/08/2026  18:45:56    <DIR>        Program Files (x86)
10/08/2026  18:45:56    <DIR>        Users
10/08/2026  18:45:56    <DIR>        Windows
10/08/2026  18:45:56     76         readme.txt

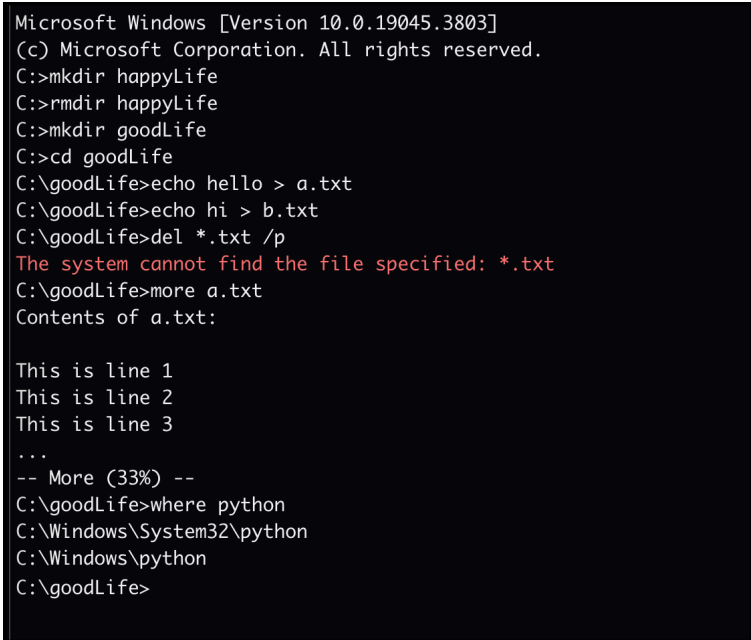
               1 File(s)      76 bytes
               6 Dir(s)
C:>cd | clip
Output copied to clipboard. (Simulator: clipboard simulated)
C:>echo My Name > test.txt
C:>type test.txt
My Name
C:>echo is Doraemon >> test.txt
```

Figure 1: Example Output

Note: Since I lack a Windows device, a online simulator has been used.

1.2 Commands in Factsheet:

```
del /p
#same /p as before but specifically for del -- confirms Y/N
↪ before deleting ech file, not force delete
exit
#closes the terminal session
dir /w
#lists contents in wide format, squeezes more names per line
↪ so less detail
dir /b
#bare format (names only), no dates/sizes/etc
echo %cd%
#prints current folder path as text
more filename
#views file page by page, pauses at each screen
python
#launches Python interpreter if installed
wgnuplot
#launches Gnuplot for plotting if installed
where python
#finds the filepath of the installed python.exe
```



```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>mkdir happyLife
C:>rmdir happyLife
C:>mkdir goodLife
C:>cd goodLife
C:\goodLife>echo hello > a.txt
C:\goodLife>echo hi > b.txt
C:\goodLife>del *.txt /p
The system cannot find the file specified: *.txt
C:\goodLife>more a.txt
Contents of a.txt:

This is line 1
This is line 2
This is line 3
...
-- More (33%) --
C:\goodLife>where python
C:\Windows\System32\python
C:\Windows\python
C:\goodLife>
```

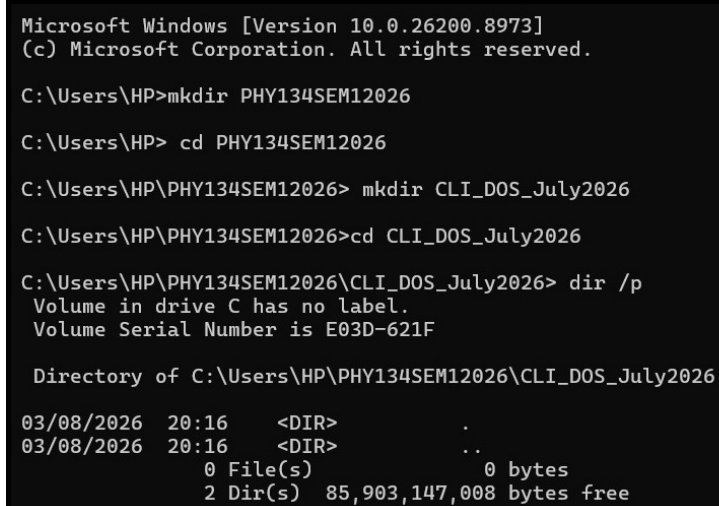
Figure 2: Sample Output on Simulator

Note: It is worth mentioning I couldn't run all the outputs on a simulator. However the tasks in section two were performed on a real Windows machine last week hence are realistic.

2. Commands and Outputs

2.1 Task 1: Directory creation and verification

```
cd PHY134SEM12026
mkdir CLI_DOS_July2026
cd CLI_DOS_July2026 dir /p
```



```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>mkdir PHY134SEM12026

C:\Users\HP> cd PHY134SEM12026

C:\Users\HP\PHY134SEM12026> mkdir CLI_DOS_July2026

C:\Users\HP\PHY134SEM12026>cd CLI_DOS_July2026

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026> dir /p
Volume in drive C has no label.
Volume Serial Number is E03D-621F

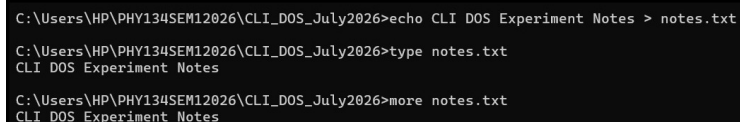
Directory of C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026

03/08/2026  20:16    <DIR>          .
03/08/2026  20:16    <DIR>          ..
               0 File(s)                0 bytes
               2 Dir(s)  85,903,147,008 bytes free
```

Figure 3: Task 1

2.2 Task 2: File creation

```
echo CLI DOS Experiment Notes > notes.txt
type notes.txt
more notes.txt
```



```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo CLI DOS Experiment Notes > notes.txt

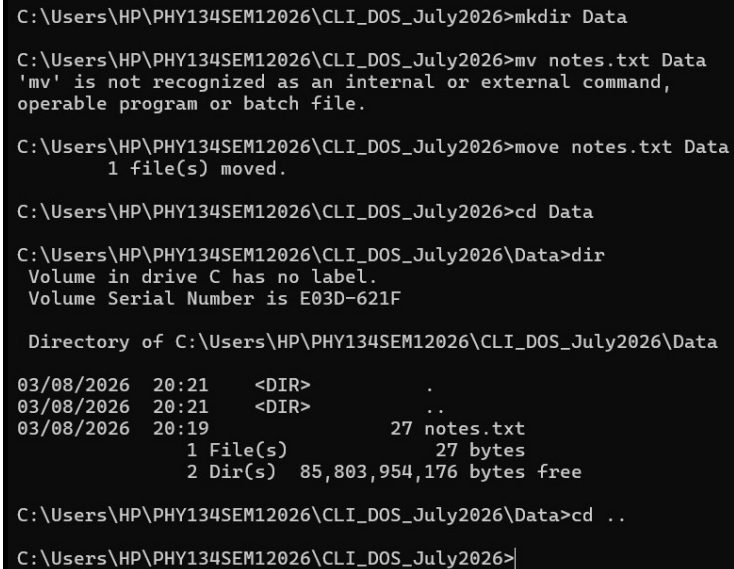
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>type notes.txt
CLI DOS Experiment Notes

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>more notes.txt
CLI DOS Experiment Notes
```

Figure 4: Task 2

2.3 Task 3: File Relocation

```
mkdir Data move notes.txt Data
cd Data dir
cd ..
```



```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>mkdir Data
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>mv notes.txt Data
'mv' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>move notes.txt Data
1 file(s) moved.
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>cd Data
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data>dir
Volume in drive C has no label.
Volume Serial Number is E03D-621F

Directory of C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data

03/08/2026  20:21    <DIR>          .
03/08/2026  20:21    <DIR>          ..
03/08/2026  20:19                27 notes.txt
               1 File(s)                27 bytes
               2 Dir(s)  85,803,954,176 bytes free

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\Data>cd ..
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>|
```

Figure 5: Task 3

2.4 Task 4: System Diagnostics Script

```
type system_report.txt
```

```

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>type system_report.txt
System Information Report
Kojagorislaptop

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2405:201:8016:e86b:57a:8104:e6c3:3f18
    Temporary IPv6 Address. . . . . : 2405:201:8016:e86b:38e8:a9e5:1e15:6717
    Link-local IPv6 Address . . . . . : fe80::2f4a:f42b:90db:7dbb%9
    IPv4 Address. . . . . : 192.168.29.167
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::b6a7:c6ff:fe4d:e95d%9
                                192.168.29.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
Name
Intel(R) Core(TM) i7-10870H CPU @ 2.20GHz
Report generated on 03/08/2026 at 20:24:41.45

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>

```

Figure 6: Task 4

2.5 Task 5: Txt File Creation and update

```

echo Test File > test.txt
echo Updated on %DATE% >> test.txt
cls

```

```

C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo Test File > test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>echo Updated on %DATE% >> test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>notepad test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>cls

```

Figure 7: Task 5

2.6 Task 6: sin(x) Plotting

```

cd gnuplot
"C:\Program Files\gnuplot\bin\gnuplot.exe"
plot sin(x)

```

```
C:\Users\HP>cd PHY134SEM12026\CLI_DOS_July2026\gnuplot
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>gnuplot
'gnuplot' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>"C:\Program Files\gnuplot\bin\gnuplot.exe"
  G N U P L O T
  Version 6.0 patchlevel 4   Last modified 2025-12-18
  Copyright (C) 1986-1993, 1998, 2004, 2007-2025
  Thomas Williams, Colin Kelley and many others
  gnuplot home:   http://www.gnuplot.info
  faq, bugs, etc: type "help FAQ"
  immediate help: type "help" (plot window: hit 'h')
  Terminal type is now qt
  Encoding set to 'cp1252'.
  gnuplot> plot sin(x)
Warning: slow font initializationgnuplot> qt_processTermEvent received a GE_fontprops event. This should not have happened
quit
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026\gnuplot>
```

Figure 8: Task 6 - Script

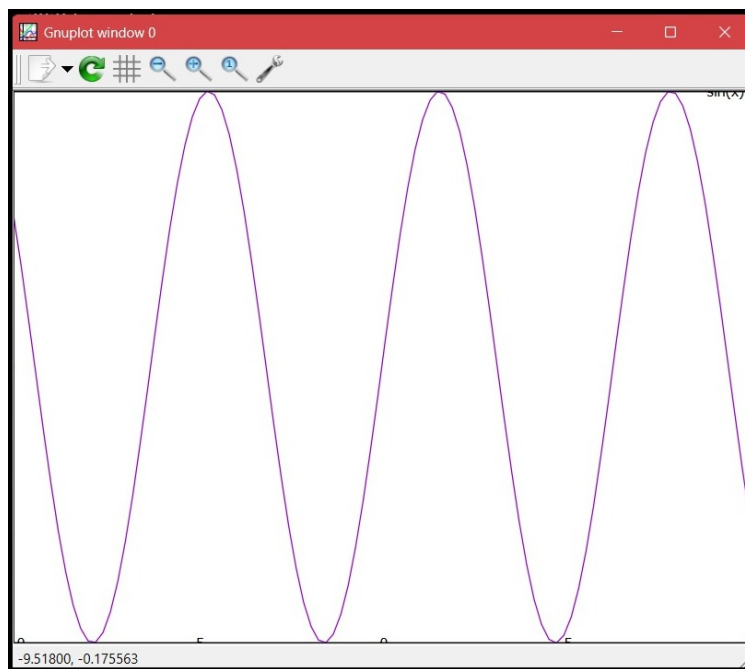
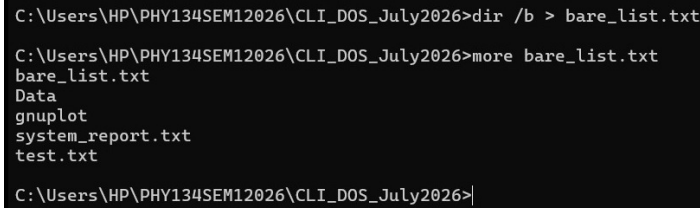


Figure 9: Task 6 - Plot

2.7 Task 7: Directory listing

```
dir /b > bare\_list.txt
more bare\_list.txt
```



```
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>dir /b > bare_list.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>more bare_list.txt
bare_list.txt
Data
gnuplot
system_report.txt
test.txt
C:\Users\HP\PHY134SEM12026\CLI_DOS_July2026>
```

Figure 10: Task 7

3. Experience (Workspace setup, MacOS)

Since I am on a machine running MacOS, the process of installation differs vastly from the one in Windows.

Firstly, the command-line package manager for MacOS isn't one distributed by Apple natively so you can use any of the free ones found online. I use brew and `brew install <package>` manages fetch/install/updates without manual intervention.

The packages notably mentioned in Lab to be installed included:

- pandoc
- Imagemagick
- GnuPlot
- Python

About Python

It is worth noting that instead of using the native `pip` globally, I prefer creating a virtual environment per project and locally install any module I need. This is a precaution against certain dependency conflicts (like different projects needing different versions) that I have faced in the past and a certain `error: externally-managed-environment` too.

About MarkText

This was the one suggested in the lab however the version for MacOS is depreciated and hence I switched to using VSCode with proper extensions. Also, worth mentioning, I used the `xelatex` renderer with pandoc since I had used that before and already had a compatible `yaml` template.

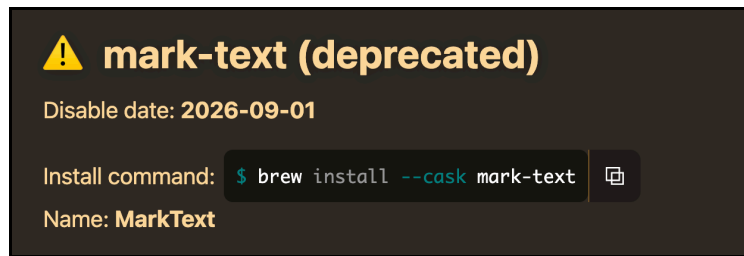


Figure 11: Depreciation Message

4. Conclusion

The advantage of CLI (vs GUI) compounds as the number of files and folders grows when in comparison, the GUI equivalent gets slower, boring and more error-prone as scale increases.

Beyond speed, the CLI (unlike GUI) allows repeating the exact same commands very easily. In MacOS, for example, you could press `Ctrl+R` and repeat any command from history, eliminating the need for retyping. This distinction matters greatly since one does not get the matter at hand successfully at first go most of the times.